Structure of the Universe I
PHYS1100-01
M-W-F from noon - 12:50 PM
245 Beacon Street, Room 107

Cyril P. Opeil, SJ, PhD
Associate Professor of Physics
Boston College Physics Department
Fall 2025

Course Description: This is an introductory course (3 credits) directed towards non-science majors. Physical principles are developed and applied to our space and astrophysical environment. Topics include the beginning, structure and evolution of our solar system, the physics of the sun, moon, and planets, space discoveries, the structure of the stars and galaxies, relativity and cosmology, extraterrestrial life, and astronomical concepts. The second half of this course (Structure of the Universe II-PHYS1101-01) will be taught in Spring 2026.

Structure of the Universe I-PHYS1100-01 and Structure of the Universe II-PHYS1100-01 are independent courses. For the best educational experience, it is desirable to take the courses in the sequence SU-I then SU-II. However, it is possible to take the courses out of sequence or just take either SU-I or SU-II.

Graduate Assistants/Homework Graders:
Several physics graduate students will be assigned to this class as teaching assistants. The class will be informed by email of their names, the location of their offices, and their office hours when this information becomes available. Graduate assistants grade exams and homework. All questions regarding the grading of homework assignments are to be directed to the graduate students. The instructor will collect the homework assignments, but will not be graded by the instructor. *All communication involving the grading of HW assignments should be directed to the teaching assistants.*

Office Hours: As posted at Higgins 330J, Monday 2:00-3:30 PM, Wednesday 2:00-3:30 PM. Most questions related to this course should be addressed in person during office hours rather than over email.

The final offices hours for the semester conclude at 3:30 PM on Monday, 08 December 2025. No offices hours will be conducted or scheduled after 3:30 PM on 08 December 2025.

Required Texts:
1) Bennett J., Donahue M., Schneider N., Voit M. (2019) *The Cosmic Perspective*, 9th Edition, Pearson Publishing Co. ISBN-10- 0134874366; ISBN-13- 978-0134874364. This book is available on Amazon and as a used book on Ebay and many other www bookstores. I have asked the BC bookstore to provide copies of this text if you wish to purchase it on campus. Electronic copies may be rented from Pearson.

This course will follow closely the text of *The Cosmic Perspective.* It is anticipated that the Structure of the Universe I - PHYS1100-01 will likely cover chapters 1-8 and the Structure of the Universe II - PHYS1101-01 will likely cover chapters 9-16. Reading the textbook prior to the class lectures will prove invaluable in comprehending the course content and assist greatly in your preparation for both mid-term and final exams.

2) Gingerich O. (2006) *God's Universe.* Belknap Press. ISBN10-0674023706; ISBN-13-978-0674023703. This book is available at the BC Bookstore, on Amazon and as a used book on Ebay and many other www bookstores.

Pre-requisites: This course is designed as an introductory survey of astronomy, cosmology and science. There are no pre-requisites for this course. A working knowledge of algebra, trigonometry and unit conversion (For example, how many inches are there in 4.35 feet?) will prove to be very helpful in this course.

Enrolling in PHYS1101-01: In the past when you enroll in PHYS1100-01 students are automatically enrolled in PHYS1101-01, Structure of the Universe II taken in Spring 2026. This is done by the Registrar's Office not by the instructor. This is to ensure students who want to continue on with the second part of the course are not excluded by other students who want to register for only the second part of this course. The course is limited to 150 students. If students do not want to take PHYS1101-01, they must remove themselves from the course by dropping PHYS1101-01. The instructor has no ability to remove students from the class roster. An instructor has the ability to permit students to enter a course roster but not remove them.

Attendance and Course Participation: Continuity and active participation are crucial to the success of this course. Students are expected to attend all lectures, participate in class discussions and exercises. Do not hesitate to ask questions in class. You are responsible for all information from each class lecture session, whether you attend or not for any reason. Excused absences due to sports activities are to be discussed with the instructor prior to the planned absence.

Attendance and participation are monitored by the Poll Everywhere App and directly effect the grades earned by students in this course. Students are required to attend class lectures to earn full marks for class participation and attendance which is 10% of your course grade.

Polling Application (app): This course will be using a polling application: *Poll Everywhere*. The software application (app) is free to download from the www to your cellphone or laptop computer. Students should expect to answer questions at random times during the lecture using this polling application. All questions will be multiple choice and students will have 45 seconds to answer each question. The results of the poll and the correct answer feedback will be given to the students immediately after the poll is complete.

Students can respond at: **PollEv.com/cyrilpopeil277**

More information on the Poll Everywhere can be found at this www address:
https://cteresources.bc.edu/documentation/poll-everywhere/

Students are responsible to register for Poll Everywhere using their BC email address. Answers provided via text message will not be recorded. Anonymous answers will not count positively towards your grade or attendance. The answers students provide to the in-class poll questions are recorded. The quality of the answers given by students to the in-class poll questions directly

affects the number of points earned in the category of: Class Participation & Polling App mentioned elsewhere in this syllabus.

If you require assistance in registering for Poll Everywhere application, discuss this with the TAs assigned to this course. Registering properly for Poll Everywhere takes only a few minutes. The information provided at: https://cteresources.bc.edu/documentation/poll-everywhere/ should answer all of your questions associated with this app.

Email: The instructor will send out HW and video assignments, announcements, *etc.* to your BC email address. It is your responsibility to check your email regularly.

Canvas: Homework assignments and important reference documents (e.g. the Syllabus) will be posted to Canvas.

Lecture Slides: The lecture slides created for and used in class lectures will not be distributed to students.

Unique Student ID Numbers for HW and Exams:
Unique student ID numbers will be assigned to all students in this class. Credit for HW assignments and exams will be assigned with this number. A spreadsheet with the unique student ID numbers and names for Fall 2025, PHYS1100-01 will be provided.

Homework:
Homework exercises will be required periodically throughout the course. The assigned questions will arrive via email with the due date for the assignment. Questions from the textbook and video assignments (available on the www) will be the subject of these homework exercises. Solutions to the problems must be clearly written out and correct to receive full credit for your solutions. You should show the equations used and outline the logic you used to obtain your solution. The graders will be instructed to take some points off for any problem that is too messy to understand. Homework assignments with no name on them will not be graded. *Homework is due at the beginning of class, on the due date noted in the scheduled assignment.*

Homework Video Availability: You can open up the YouTube videos in web browsers like Chrome, Safari, Firefox or DuckDuckGo by clicking on the link provided in the homework assignment. If you use the browser DuckDuckGo you will not have to view advertisements that pop-up during most YouTube videos.

Homework Sorting Pages:
To assist the TAs in correcting the volume of HW assignments, sorting pages will be required for submitting HW assignments. The sorting page should be the first page of your homework assignment are submitting. On the sorting page indicate the HW assignment number you are submitting and write your unique student ID# in the space provided. Sorting bins corresponding to your unique student ID# will be provided. The bin numbers are: **1000-1099, 1100-1199, 1200-1299, 1300-1399… 1900-1999**. Copies of those "sorting pages" will be available in class or may be printed out by a student.

All questions and communication involving HW assignments should be directed to the teaching assistants. The Instructor/Lecturer will not be grading the homework assignments, midterm exams or the final exam part II.

Students can pick up the corrected HW assignments after class or during office hours.

No HW assignment will be graded for credit if received after the date/time of the final class which is 08 December 2025 @ 1:00 PM. There are no exceptions.

Midterm Exam:
There will be one 50-minute midterm exam scheduled half way through the course, see the schedule below. If a question is multiple choice, A, B, C…, then you may only select one of the possible answers. Clearly, indicate your choice. Any problem solutions must be clearly written out and correct to receive full credit for your answer. You should show the equations used and outline the logic you used to obtain your solution. Materials included in the midterm exam will pertain to the lecture materials, homework assignments and our textbook. Any guest lecture presentations or video assignments will not be part of the midterm. The grader will be instructed to take some points off for any problem that is too messy to understand. Midterm make-up exams will only be made for exceptional reasons requiring previous approval of the instructor, and/or a note from the Dean's office and it is subject to the instructor's discretion. **

Final Exam Part I:
An essay question: Owen Gingerich discusses Intelligent Design and intelligent design in his book *God's Universe:* (Note the capital and lowercase letters… they are important.)
i. Briefly explain what is the difference between Intelligent Design and intelligent design as understood by Gingerich. (1 page maximum)
ii. What does Gingerich say in *God's Universe* about intelligent design? (3 pages maximum)
iii. Explain why do you agree or disagree with Gingerich's position on intelligent design and Intelligent Design? (4 pages maximum)
.
Minimum & Maximum Page Count: What is the minimum number of pages for the parts of this essay assignment? Students should be aware that a response to such profound questions may be challenging to answer in just a few pages. Due credit will be given for essays that are briefer than the maximum number of pages. (In the parlance of baseball, do not bunt, but swing for the fence.)

This assignment is an essay not a research paper. Students must read and understand what Gingerich is saying in his book *God's Universe.* This assignment is designed to make students think, reflect and understand a fundamental question in contemporary Cosmology. Other reference works and citations are not forbidden but tend to add little to the substance of the required essay. Reading other resources about this essay topic are to be encouraged. Essays that address "intelligent design" from perspectives other than the one Owen Gingerich poses likely will not answer the requirements of the essay.

QuillBot: There is a free artificial intelligence application (AI app) available to review the grammar, spelling and punctuation of your essay that you can download from the web. This app

can be extremely useful in correcting any English grammar errors in your essay. Using the QuillBot app to actually create or write your essay is not permitted. Helping to improve the quality of your submission by using QuillBott is permitted and strongly encouraged.

Essay Format: Endnotes or footnotes are required for any references (cited works) in the essay. The last page should include a list all references used in the essay. The page containing the references is not included in page count maximum. The paper should double spaced, and utilize Times New Roman or Arial 12-point font with one inch border margins. An example of proper formatting will be available to students.

Essay Delivery: Send your completed essay as an attachment in pdf format to my email address: opeil@bc.edu. The unique file name of your essay sent to the instructor should be "Gingerich Essay Your Name.pdf"; if your name is Abraham Lincoln, then the file name would be "Gingerich Essay Abraham Lincoln.pdf". Do not send the essay using Google Drive, private bicycle courier, carrier pigeon or any other method or software (Word [.doc] or text document [.txt]) to deliver your attachment. Essays are due by the date noted in the course schedule 21 November 2025 at 5:00 PM. **Essays will not be accepted for credit after the deadline**. The use of GTP Chat or other AI chatbot technology resources are not permitted in creating or writing this essay. Corrected essays will be evaluated by the instructor and returned to the students on the day of the final exam.

Essay Check List:
1) The document format of the essay that was attached and sent to opeil@bc.edu was a pdf? _____
2) Did the essay list its references on the last page? ____
3) Is the format of the essay similar to the example provided by the instructor? ____
4) Was the essay was sent to opeil@bc.edu by the due date and time noted in the course schedule 21 November 2025 at 5:00 PM. ____
5) The essay was limited to eight pages plus the references. ____
6) GTP Chat or other AI chatbot technology resources were not used in producing the essay text? ____
7) The unique file name of your essay sent to the instructor should be in the form "Gingerich Essay Your Name.pdf"; if your name is Abraham Lincoln, then the file name would be "Gingerich Essay Abraham Lincoln.pdf". ____

Final Exam Part II:
There will be an in-class written final exam on the date and time assigned to this class by Boston College. The problems must be clearly written out and correct to receive full credit for your solutions. You should show the equations used and outline the logic you used to obtain your solution. The grader will be instructed to take some points off for any problem that is too messy to understand.

Final exam schedule issues (such as having three exams in one day, or a direct conflict) should be reported to the instructor on or before 01 December 2025. A make-up final exam can be scheduled for a date after the scheduled final exam for those students who have verified conflicts and have notified the instructor before this deadline. The instructor will not schedule a separate

make-up exam to accommodate travel arrangements; it is the student's responsibility to schedule their travel plans to accommodate their own final exam schedule set by the university.**

Final make-up exams will only be made for exceptional reasons requiring previous approval of the Instructor, and/or a note from the Dean's office and it is subject to the instructor's discretion.**

**If you are ill or have some other issue for missing a homework assignment, a planned midterm date, or the final exam date, plan to discuss this with the instructor in person. Such a request may need a letter from a doctor/nurse practitioner, or from an official on the campus.

Grade Assessments:
Grades for this course are assessed and calculated after the final exam is evaluated. Grades will not be calculated or assessed prior to the completion of the final exam. If a student wants to know how they are doing in the course, they need to know how many points they have received for the midterm exam and each homework assignment. Students are encouraged to calculate their own grades and know how they are doing in the course. Class participation is evaluated by Poll Everywhere and can only be assessed after the end of the last class. Students need to collect their corrected HW assignments and exams to know how they are doing in the course. The syllabus clearly lists how HW assignments and exams are weighted in determining a final grade for the course.

Requesting an Incomplete: Boston College Academic Policy states: All required work in any course must be completed by the date set for the course examination. A student who has not completed the exams, research or written work for a course taken in the fall or spring semester or is absent from the course examination in either semester can, with adequate reason and at the discretion of the instructor, receive a temporary grade of Incomplete (I). All such "I" grades will automatically be changed to "F" on March 1 for the fall, August 1 for the spring, and October 1 for the summer. After that time, no further grade change will be allowed.

Requesting an incomplete grade for any course is an extraordinary request. Students may request an incomplete grade in this course, if and only if the student, for serious reasons, cannot complete the requirements of the course by the date of the final exam. All requirements of this course must be completed by the date set by the instructor. Dates for final exam are set by the university. Incomplete grade requests are generally granted by the instructor in the event of an accident that disables the student from completing the work by the date of the final, for physical or mental health reasons. The convenience and travel schedules of students are not sufficient to request an incomplete. If you are considering asking the instructor for an incomplete in this course, come to the office hours provided by the instructor to discuss your request. The MCAS Dean's Office does not grant requests for an incomplete in a course, instructors do. In point of fact, the MCAS Dean explicitly discourages granting Incomplete, "I" grades at the end of a course.

Students who do not attend the midterm or final exam and have made no provision to take it at another time/date will receive 0/25 points on the exam. The essay is worth 0/20 points if not received by the time/date listed in the syllabus.

Grading:	Class Participation, Poll Everywhere	10/10 points
	Homework	20/20 points
	Mid-term Exam	25/25 points
	Essay	20/20 points
	Final Exam	25/25 points

Eating and Drinking: *Absolutely* no food or beverages are to be brought into or consumed in the lecture room. Not consuming of food and or beverages in the lecture hall is a matter of simple courtesy and civility toward the instructor and your fellow students.

Cell Phone Use: Students are required to turn off their cell phone ringtone prior to attending class. It is a matter of courtesy to the instructor and other students that cell phones do not ring during class time.

Zoom: Students should have the access, knowledge and ability to use Zoom. Lectures during this class can be delivered via Zoom to the students by the instructor or an outside lecturer. If a Zoom lecture is scheduled, students to not have to come to the classroom, but they do have to attend the lecture.

Computer Use During Lectures: Students are asked not to use personal computers during the lecture times. Listen, take notes with a pen or pencil and ask questions during the lecture. The temptation to use your available technology (laptops, iPhones etc.) for other web content (emails, social media etc.) often conflicts with a student's ability to listen to and comprehend the lecture presentations.

Academic Integrity: Discussion of the concepts and problem-solving techniques between students, and with the instructor and TA is strongly encouraged. However, under no circumstances is it acceptable to copy and turn in solutions taken from ANY source. Please review the Boston College Standards of Academic Integrity:

https://www.bc.edu/bc-web/academics/sites/university-catalog/policies-procedures.html

It is your responsibility to ensure that all work (exams, essay and homework) you submit is in accordance with university policies. If you have any questions, please consult with the instructor. Violations will be reported to the Deans' Office and reviewed by the College's Committee on Academic Integrity. Violations could result in failure in the course and administrative sanctions e.g. probation or suspension.

Office of Health Promotion: If you are feeling stressed, having challenges managing your time, sleep, or making choices around alcohol and food, the Office of Health Promotion offers Wellness Coaching appointments to support your health and wellbeing. Please reach out by going to the Office of Health Promotion shown below to schedule a virtual meeting with a staff member and Wellness Coach and for health and wellness information.

https://www.bc.edu/bc-web/offices/student-affairs/sites/center-for-student-wellness/about/anniversary.html

Disability Accommodation: If you have a disability and will be requesting a disability accommodation for this course, please inform the instructor. Information concerning disability accommodation can be found by contacting Paulette Durrett, the Assistant Dean for Students with Disabilities (durrett@bc.edu) 617-552-3470. Boston College provides, through the Connors Family Learning Center, a facility to assist people obtain disability accommodations. Further information can be found on the www link:

https://www.bc.edu/content/bc-web/academics/sites/connors-family-learning-center/services/ADHD-Learning-Disability-Support-Services.html

Title IX Harassment Policy: Boston College students, employees, coaches, and instructors are subject to comply with the current Harassment Policy. Violations of this policy are rare but are in no way tolerated by members of the Boston College community. If you have any questions about the scope of the Boston College Harassment Policy and penalties, please refer to:

https://www.bc.edu/content/dam/bc1/sites/policies/Title-IX-Harassment-Policy.pdf

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Schedule of Classes:

Monday, 25 August: Lecture 1 – Introduction, Syllabus, Why does BC teach this course? Jesuits and Science.

Homework Assigned: Reads from Genesis, Wisdom & Qur'an

Wednesday, 27 August: Lecture 2 – A look at Genesis, Wisdom & Qur'an; Aristotle's Four Causes & Chap. 1: A Modern View of the Universe

Friday, 29 August: Lecture 3

Monday, 01 September: No class, Labor Day

Wednesday, 03 September: Lecture 4

Friday, 05 September: Lecture 5 - HW 1 due in class

Monday, 08 September: Lecture 6

Wednesday, 10 September: Lecture 7

Friday, 12 September: Lecture 8 - HW 2 due in class

Monday, 15 September: Lecture 9

Wednesday, 17 September: Lecture 10

Friday, 19 September: Lecture 11

Monday, 22 September: Lecture 12

Wednesday, 24 September: Lecture 13 - HW 3 due in class

Friday, 26 September: Lecture 14

Monday, 29 September: Lecture 15

Wednesday, 01 October: Lecture 16

Friday, 03 October: Lecture 17

Monday, 06 October: Lecture 18 - HW 4 due in class

Wednesday, 08 October: Lecture 19

Friday, 10 October: Lecture 20

Monday, 13 October: No class, Fall Break

Wednesday, 15 October: Lecture 21
Friday, 17 October: Lecture 22
Monday, 20 October: Midterm Exam
Wednesday, 22 October: Lecture 23 - HW 5 due in class
Friday, 24 October: Lecture 24
Monday, 27 October: Lecture 25
Wednesday, 29 October: Lecture 26
Friday, 31 October: Lecture 27
Monday, 03 November: Lecture 28
Wednesday, 05 November: Lecture 29 - HW 6 due in class
Friday, 07 November: Lecture 30
Monday, 10 November: Lecture 31
Wednesday, 12 November: Lecture 32
Friday, 14 November: Lecture 33
Monday, 17 November: Lecture 34 - HW 7 due in class
Wednesday, 19 November: Lecture 35
Friday, 21 November: Lecture 36, Essay in pdf format due by 5:00 PM by email to opeil@bc.edu
Monday, 24 November: Lecture 37
Wednesday, 26 November: No class, Thanksgiving Holiday
Friday, 28 November: No class, Thanksgiving Holiday
Monday, 01 December: Lecture 38
Wednesday, 03 December: Lecture 39 - HW 8 due in class
Friday, 05 December: Lecture 40
Monday, 08 December: Lecture 41, Last day of classes for Fall semester. Offices Hours from 2:00-3:30 PM for the semester conclude at 3:30 PM. No offices hours will be conducted after 3:30 PM.
Final Examination for PHYS1100-01, will be held from 9:00 -11:59 AM, 245 Beacon, Room 107. The date has not yet been set by the University, but will be announced to the class when the date is set.

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PHYS1100: Structure of the Universe I
Homework, Essay Assignments & Midterm due dates - Fall 2025
Some HW due dates may be adjusted as the semester progresses.

Friday, 05 September: HW 1 due in class
Friday, 12 September: HW 2 due in class
Wednesday, 24 September: HW 3 due in class
Monday, 06 October: HW 4 due in class
Monday, 20 October: Midterm Exam
Wednesday, 22 October: HW 5 due in class
Wednesday, 05 November: HW 6 due in class
Monday, 17 November: HW 7 due in class
Friday, 21 November: Essay in pdf format due by 5:00 PM by email to opeil@bc.edu
Wednesday, 03 December: HW 8 due in class
Final exam some time after 08 December 2025, date & time to be determined.

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"All I'm saying is now is the time to develop the technology to deflect an asteroid."